INFOSYS SE REASONING VAULT • CRITICAL REASONING & STATEMENT LOGIC

Statement & Assumptions, Course of Action, Strong/Weak Arguments • Total Solved Questions: 38

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q1. Statement: Should school uniforms be made compulsory for students? Arguments: I. Yes, uniforms create equality among students and remove discrimination based on clothing. II. No, students should have the freedom to wear what they want.

(A) Only Argument I is strong
(B) Only Argument II is strong
(C) Both I and II are strong
(D) Neither I nor II is strong

Correct Answer: Option (A) : Only Argument I is strong

■ **Step-by-Step Logic:** Argument I is strong as it supports discipline and equality. Argument II is weak as it focuses only on personal freedom without considering the benefits of uniforms.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q2. Statements: 1. The electricity demand has sharply increased, leading to frequent power outages. 2. The city has experienced its hottest summer in the last 50 years.

(A) Statement 1 is the cause, and Statement 2 is its effect.
(B) Statement 2 is the cause, and Statement 1 is its effect.
(C) Both statements are independent causes.
(D) Both statements are effects of a common cause.

Correct Answer: Option (B) : Statement 2 is the cause, and Statement 1 is its effect.

■ **Step-by-Step Logic:** The extreme heat (Statement 2) caused an increase in electricity usage (Statement 1), leading to power outages.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q3. Statements: 1. Many areas in the city are facing severe water shortages. 2. The monsoon rainfall this year was significantly lower than expected.

(A) Statement 1 is the cause, and Statement 2 is its effect.
(B) Statement 2 is the cause, and Statement 1 is its effect.
(C) Both statements are independent causes.
(D) Both statements are effects of a common cause.

Correct Answer: Option (B) : Statement 2 is the cause, and Statement 1 is its effect.

■ **Step-by-Step Logic:** Low monsoon rainfall (Statement 2) has led to severe water shortages (Statement 1), making Statement 2 the cause and Statement 1 the effect.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q4. Statement: Should social media platforms be held legally responsible for spreading fake news? Arguments: I. Yes, because social media plays a significant role in shaping public opinion and should be accountable. II. No, because it is impossible to monitor and control every piece of information shared by users.

(A) Only Argument I is strong
(B) Only Argument II is strong
(C) Both I and II are strong
(D) Neither I nor II is strong

Correct Answer: Option (C) : Both I and II are strong

■ **Step-by-Step Logic:** Both arguments are strong. Social media influences public perception (I), but monitoring all content is impractical (II).

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q5. Statement: "Employees who do not complete their training within three months will be terminated from their jobs." – A company's HR policy. Conclusions: I. Completing training is necessary to continue employment in this company. II. Some employees fail to complete their training within three months.

(A) Only Conclusion I follows
(B) Only Conclusion II follows
(C) Both I and II follow
(D) Neither I nor II follows

Correct Answer: Option (A) : Only Conclusion I follows

■ **Step-by-Step Logic:** I follows because the statement makes training mandatory. II does not follow since the statement does not confirm that employees have failed the training, only that they will be terminated if they do.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q6. Argument: Wearing a seatbelt while driving reduces the risk of injury in accidents.

(A) Many people find seatbelts uncomfortable.
(B) Seatbelts are mandatory in many countries.
(C) Seatbelts prevent passengers from being thrown forward in case of a sudden stop or crash.
(D) Accidents can also be caused by reckless driving.

Correct Answer: Option (C) : Seatbelts prevent passengers from being thrown forward in case of a sudden stop or crash.

■ **Step-by-Step Logic:** Seatbelts prevent passengers from being thrown forward, which directly supports the argument that they reduce the risk of injury in accidents.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q7. Argument: Reading books enhances knowledge and improves vocabulary.

(A) Watching movies is a good way to relax.
(B) Many people do not enjoy reading.
(C) Reading books takes a lot of time.
(D) Books contain information that helps people learn new words and concepts.

Correct Answer: Option (D) : Books contain information that helps people learn new words and concepts.

■ **Step-by-Step Logic:** Books provide knowledge and introduce new vocabulary, which directly strengthens the argument.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q8. Statement: Many factories have been dumping industrial waste into a major river, causing severe pollution. Courses of Action: I. The government should shut down all factories near the river. II. Strict penalties should be imposed on factories violating pollution norms.

(A) Only Course of Action I follows
(B) Only Course of Action II follows
(C) Both I and II follow
(D) Neither I nor II follows

Correct Answer: Option (D) : Neither I nor II follows

■ **Step-by-Step Logic:** Shutting down all factories (Course I) is an extreme measure that could harm the economy. Simply imposing penalties (Course II) may not be enough without proper enforcement. A better solution would be stricter monitoring and implementation of waste treatment systems.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q9. Statement: Many students in a school are failing in mathematics. Courses of Action: I. The school should introduce extra coaching classes for mathematics. II. The teaching methods for mathematics should be reviewed and improved.

(A) Only Course of Action I follows
(B) Only Course of Action II follows
(C) Both I and II follow
(D) Neither I nor II follows

Correct Answer: Option (C) : Both I and II follow

■ **Step-by-Step Logic:** Both actions are reasonable. Extra coaching will provide immediate support, while improving teaching methods will have long-term benefits.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q10. Statement: "Many people are shifting to electric vehicles due to rising fuel prices." Conclusion: I. Rising fuel prices is not affecting people's choice of transportation. II. All people will eventually switch to electric vehicles.

(A) Only Conclusion I follows
(B) Only Conclusion II follows
(C) Both I and II follow
(D) Neither I nor II follows

Correct Answer: Option (D) : Neither I nor II follows

■ **Step-by-Step Logic:** Conclusion I do not follow because the statement does not establish a cause-effect relationship. Conclusion II does not follow as it is an extreme generalization.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q11. Statement: Should the government implement a strict ban on plastic bags? Arguments: I. Yes, plastic bags contribute to environmental pollution and should be banned. II. No, banning plastic bags will cause inconvenience to consumers and businesses.

(A) Only Argument I is strong
(B) Only Argument II is strong
(C) Both I and II are strong
(D) Neither I nor II is strong

Correct Answer: Option (A) : Only Argument I is strong

■ **Step-by-Step Logic:** Argument I is strong because it presents an environmental concern. Argument II is weak because inconvenience alone is not a strong enough reason to ignore environmental damage.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q12. Read the following argument and mark which of the following assumptions is made in the argument. More than one billion years' worth of rocks have gone missing from the geologic record of Grand Canyon.

(A) Earth has transitioned from an older setting to the modern one during these one billion years.
(B) The western half of the Grand Canyon has gone through a very different geologic contortion compared to the eastern half.
(C) Series of Faulting events in that region possibly tore up the earth around the Canyon causing rocks and sediments to wash away in the ocean.
(D) None of the mentioned options.

Correct Answer: Option (C) : Series of Faulting events in that region possibly tore up the earth around the Canyon causing rocks and sediments to wash away in the ocean.

■ **Step-by-Step Logic:** In a new study, researchers led by Peak identify a likely cause for the missing rocks: the breakup of Earth's ancient supercontinent Rodina approximately 700 million years ago. The upheaval was so violent it likely washed rocks and sediment into the ocean — one billion years' worth, in the case of the Grand Canyon.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q13. Statement: Fluency in spoken English is a must have in today's job market, where every firm wants to compete at the international level. Inferences: I. The imposition of language is unfair. II. English is the language that bridges the gap between countries.

(A) Only Conclusion I follows
(B) Only Conclusion II follows
(C) Both I and II follow
(D) Neither I nor II follows

Correct Answer: Option (B) : Only Conclusion II follows

■ **Step-by-Step Logic:** Conclusion I: "The imposition of language is unfair" does not follow because the statement does not discuss fairness or imposition; it simply states a requirement in the job market. Conclusion II: "English is the language that bridges the gap between countries" follows because the statement implies that fluency in English is important for firms competing internationally.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q14. Statements: 1. The company has launched a new marketing campaign with attractive discounts. 2. The sales of the company have significantly increased in the past month.

(A) Statement 1 is the cause, and Statement 2 is its effect.
(B) Statement 2 is the cause, and Statement 1 is its effect.
(C) Both statements are independent causes.
(D) Both statements are effects of a common cause.

Correct Answer: Option (A) : Statement 1 is the cause, and Statement 2 is its effect.

■ **Step-by-Step Logic:** The launch of a new marketing campaign (Statement 1) likely led to an increase in sales (Statement 2), making it a cause-and-effect relationship.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q15. Directions: In the question below, a statement and assumptions numbered I and II are given. An assumption is something supposed or taken for granted. You have to consider the following assumption and decide which of the assumption is implicit in the statement. Statement: "Tomorrow morning the daily review meeting will happen at 1 pm instead of 11 am" said the boss. Assumptions: I. That day boss will arrive at the office only at 1 pm. II. Some of the employees will not be available at 11 am.

(A) Only Assumption I is implicit
(B) Only Assumption II is implicit
(C) Both I and II are implicit
(D) Neither I nor II is implicit

Correct Answer: Option (D) : Neither I nor II is implicit

■ **Step-by-Step Logic:** The statement does not mention when the boss will arrive (I is not implicit). The reason for rescheduling is unclear, so employee unavailability is not necessarily assumed (II is not implicit).

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q16. Statement: Mother scolding the child said, "If you continue doing naughty things, I will slap you." Assumptions: I. All children are naughty. II. With this warning, child may stop doing mischievous acts.

(A) Only Assumption I is implicit
(B) Only Assumption II is implicit
(C) Both I and II are implicit
(D) Neither I nor II is implicit

Correct Answer: Option (B) : Only Assumption II is implicit

■ **Step-by-Step Logic:** Assumption I: "All children are naughty" is not implicit because the statement talks about one specific child, not all children in general. Assumption II: "With this warning, the child may stop doing mischievous acts" is implicit because the mother's warning suggests that she believes it will have an effect on the child's behavior.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q17. Social gatherings could turn into COVID super spreader events if safety norms are flouted. Conclusions : I. The coming festive season could be dangerous if safety norms are flouted and lead to a surge in COVID cases. II. Vaccine provides over 95% protection against severe disease and hospitalisation.

(A) Only Conclusion I follows
(B) Only Conclusion II follows
(C) Both I and II follow
(D) Neither I nor II follows

Correct Answer: Option (A) : Only Conclusion I follows

■ **Step-by-Step Logic:** As per the statement gatherings could turn in to COVID super spread , so only Conclusion I follows, decision on Conclusion II can't be taken as the statement doesn't discuss about vaccination.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q18. Statement: "People who exercise regularly are healthier than those who don't." Assumption: I. Regular exercise improves health. II. Some people do not exercise regularly.

(A) Only Assumption I is implicit
(B) Only Assumption II is implicit
(C) Both I and II are implicit
(D) Neither I nor II is implicit

Correct Answer: Option (C) : Both I and II are implicit

■ **Step-by-Step Logic:** Assumption I is implicit because the statement claims a direct relationship between exercise and health. Assumption II is also implicit as the comparison suggests the existence of people who do not exercise.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q19. Statement: "All cats are mammals. Some mammals are not domestic animals." Conclusion: I. Some domestic animals are mammals. II. All mammals are cats.

(A) Only Conclusion I follows
(B) Only Conclusion II follows
(C) Both I and II follow
(D) Neither I nor II follows

Correct Answer: Option (A) : Only Conclusion I follows

■ **Step-by-Step Logic:** Conclusion I logically follows, as some mammals can be domestic. Conclusion II is incorrect because all mammals are not necessarily cats.

[OnlineStudy4U Topicwise Series (Critical Reasoning)]

Topic: Critical Reasoning

Q20. Statement: "To get a driving license, you must pass a driving test." Assumption: I. Driving tests help in identifying competent drivers. II. Some people fail the driving test.

(A) Only Assumption I is implicit
(B) Only Assumption II is implicit
(C) Both I and II are implicit
(D) Neither I nor II is implicit

Correct Answer: Option (C) : Both I and II are implicit

■ **Step-by-Step Logic:** Assumption I is implicit because the requirement of the test suggests its importance. Assumption II is implicit because if everyone passed, there would be no need for a test.

[College ITM Mock Paper 1]

Topic: Reasoning Ability

Q21. A solid cube painted red on all faces is cut into 64 identical small cubes (4 × 4 × 4). How many small cubes have exactly two faces painted?

(A) 12
(B) 24
(C) 36
(D) 8

Correct Answer: Option (A) : 12

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (A) 12. All condition checks confirm consistency.

[College ITM Mock Paper 1]

Topic: Reasoning Ability

Q22. In triangle ABC, the medians AD, BE and CF are drawn and meet at G. How many triangles are there in the figure in total?

(A) 12
(B) 14
(C) 16
(D) 18

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 1]

Topic: Reasoning Ability

Q23. Find the next letter: B, E, I, N, T, ?

(A) A
(B) Z
(C) B
(D) Y

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 2]

Topic: Reasoning Ability

Q24. In a code, every letter is shifted 3 places forward in the alphabet (wrapping Z→C), so FLOWER is written as IORZHU. Using the same code, how is GARDEN written?

(A) JDUGHQ
(B) JDUFHQ
(C) JCUGHQ
(D) JDVGHQ

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 2]

Topic: Reasoning Ability

Q25. A cube is painted red on one pair of opposite faces, blue on another pair, and green on the last pair. It is then cut into 27 identical small cubes. How many small cubes have exactly one face painted?

(A) 6
(B) 8
(C) 9
(D) 12

Correct Answer: Option (A) : 6

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (A) 6. All condition checks confirm consistency.

[College ITM Mock Paper 2]

Topic: Reasoning Ability

Q26. A rectangle is divided into a 3 × 3 grid of smaller cells by 2 evenly spaced vertical lines and 2 evenly spaced horizontal lines drawn inside it. How many rectangles (of any size) are there in the whole figure?

(A) 30
(B) 36
(C) 27
(D) 24

Correct Answer: Option (A) : 30

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (A) 30. All condition checks confirm consistency.

[College ITM Mock Paper 2]
Topic: Reasoning Ability

Q27. Find the next letter-pair: CE, FI, IM, LQ, ?

(A) OT
(B) OU
(C) NT
(D) PU

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 3]
Topic: Reasoning Ability

Q28. A cuboid measuring 5 × 4 × 3 units is painted on all its outer surfaces and then cut into unit cubes. How many unit cubes have exactly 2 painted faces?

(A) 24
(B) 22
(C) 20
(D) 26

Correct Answer: Option (A) : 24

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (A) 24. All condition checks confirm consistency.

[College ITM Mock Paper 3]
Topic: Reasoning Ability

Q29. How many triangles are there in total in a complete five-pointed star (pentagram) figure?

(A) 5
(B) 10
(C) 35
(D) 15

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 4]
Topic: Reasoning Ability

Q30. How many squares (of all sizes) are there in a 4×4 grid of unit squares?

(A) 30
(B) 25
(C) 36
(D) 20

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 5]
Topic: Reasoning Ability

Q31. A rectangular cuboid is cut by 2 parallel cuts along its length, 1 cut along its breadth, and 3 cuts along its height (each cut perpendicular to that dimension), producing smaller cuboids from an originally fully-painted surface. How many of the resulting small cuboids have exactly 1 painted face?

(A) 4
(B) 6
(C) 8
(D) 2

Correct Answer: Option (A) : 4

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (A) 4. All condition checks confirm consistency.

[College ITM Mock Paper 5]
Topic: Reasoning Ability

Q32. How many rectangles (of all sizes, including squares) are there in a 4×4 grid of unit squares?

(A) 100
(B) 90
(C) 81
(D) 110

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 6]
Topic: Reasoning Ability

Q33. In a code, the value of a word = (sum of the alphabetical positions of its letters × number of letters) – (number of vowels in the word). Given that CAB has the value 17, what is the value of DIGIT?

(A) 243
(B) 245
(C) 241
(D) 247

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 6]
Topic: Reasoning Ability

Q34. A solid cube of side 7 units is painted on all faces and cut into 1-unit cubes. How many small cubes have at most 1 painted face?

(A) 275
(B) 125
(C) 150
(D) 335

Correct Answer: Option (A) : 275

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (A) 275. All condition checks confirm consistency.

[College ITM Mock Paper 6]
Topic: Reasoning Ability

Q35. How many rectangles that are NOT squares can be found in a 4 × 5 grid of unit squares?

(A) 110
(B) 150
(C) 100
(D) 120

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 6]
Topic: Reasoning Ability

Q36. Five people A, B, C, D and E stand in a queue. A is ahead of B but behind C. D stands immediately behind B. E is ahead of C. Who stands third from the front?

(A) A
(B) B
(C) C
(D) E

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 6]

Topic: *Reasoning Ability*

Q37. Find the next group: ACF, GIL, MOR, ?

(A) SUX
(B) SVY
(C) TVY
(D) SUW

Correct Answer: See Logic

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option () . All condition checks confirm consistency.

[College ITM Mock Paper 6]

Topic: *Reasoning Ability*

Q38. A square sheet of paper is folded in half three times (each fold halves the previous folded shape). A single hole is then punched through all layers. When the sheet is fully unfolded, how many holes does it have?

(A) 8
(B) 4
(C) 6
(D) 16

Correct Answer: Option (A) : 8

■ **Step-by-Step Logic:** Logical Deduction: Evaluating the given premises and constraints step-by-step establishes that the verified outcome is Option (A) 8. All condition checks confirm consistency.